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Full Length Article

Discrete bidirectional memristive neural network-based hyperchaotic system and its FPGA implementation

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ABSTRACT

Memristors, owing to their unique non-volatile and nonlinear characteristics, have been widely used to construct biomimetic neural network models. In recent years, discrete-time memristive neural networks (DMNN) have demonstrated significant research value and application potential in the field of chaotic dynamics. However, in existing designs on DMNN-based chaotic systems, there are no relevant reports on the use of novel neural network models with bidirectional feedback structures. In this paper, we propose a hyperchaotic system based on a discrete bidirectional memristive neural network (DBMNN) and implement it on an FPGA platform. A discrete locally active memristor is embedded into the feedback path of a discrete bidirectional neural network (DBNN), forming a six-dimensional DBMNN. The dynamical behavior of DBMNN is theoretically and experimentally studied using various dynamical analysis methods, including equilibrium point analysis, phase diagrams, Lyapunov exponent analysis, bifurcation diagrams, and bi-parameter dynamic maps. The results show that the system exhibits complex chaotic characteristics, including hyperchaos with four positive Lyapunov exponents, heterogeneous coexisting attractors, transient chaos, and chaotic attractor jumping. A large range of amplitude control phenomena is also found. Finally, the circuit design of DBMNN is implemented on a field-programmable gate array (FPGA), and the experimental results are consistent with the numerical simulations, verifying the dynamics of the proposed hyperchaotic system.

1. Introduction

The human brain's neural network, comprising tens of billions of interconnected neurons, constitutes a highly complex nonlinear dynamical system that has been proven to exhibit chaotic behavior (Korn & Faure, 2003). Numerous studies have revealed that the brain's chaotic dynamics are closely related to cognitive functions such as thinking, memory, and learning (Ma & Tang, 2017; Suárez et al., 2021), which play a crucial role in the field of artificial intelligence. In order to further explore the dynamical properties of the brain, researchers have developed brain-like chaotic systems based on various artificial neural network models, including Hopfield neural network (HNN), cellular neural networks (CNN), Hindmarsh-Rose (HR) neuronal network, and FitzHugh-Nagumo (FHN) neuronal network (Etémé et al., 2020; Huang et al., 2012; Santana et al., 2021; Xu et al., 2022). Moreover, these neural network-based chaotic systems also have application value in areas such as image encryption (Xu et al., 2022), secure communications (Subramaniam & Mani, 2024), and biomedicine (Panahi et al., 2017). Therefore, research on the chaotic dynamics of neural networks holds substantial importance both theoretically and practically.

Recent advances in neuromorphic platforms have underscored the significance of locally active memristive devices, high-precision synaptic elements, and biomimetic nonlinear dynamics for functional neural systems (Wu et al., 2025; Yang et al., 2025a,b). For brain-like chaos, memristive neural networks(MNNs) have gradually emerged as a research hotspot in academia due to their exceptional biomimetic characteristics (Fan & Wang, 2025). The theoretical model of the memristor was first proposed by Chua in 1971 to describe the relationship between charge and magnetic flux (Chua, 1971). In 2008, HP Labs developed a nano-memristor device from a physical perspective (Strukov et al., 2008), which triggered a boom in memristor research that has led to widespread utilization of memristors in simulating the dynamics of neurons and neural networks (Deng et al., 2025d; Tian et al., 2025a; Wang et al., 2025). As a new type of electronic component, the memristor possesses numerous excellent characteristics, including nonlinearity, non-volatility, and local activity. It can emulate synaptic connections

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between neurons or represent the influence of electromagnetic effects on neural behavior, and this critical biomimetic functionality allows memristors to replace conventional resistors in constructing neural networks (Luo et al., 2025b; Yu et al., 2024b). On one hand, novel chaotic systems constructed on MNN more closely approximate real neural systems, being capable of exhibit hyperchaos (Li et al., 2014), transient chaos (Hu et al., 2018), coexisting attractors (Lai & Guo, 2024), hidden attractors (Pham et al., 2016), multistability (Ding et al., 2022), and multi-ring/multi-scroll behaviors (Deng et al., 2025c; Kong et al., 2024). Chaos phenomena are increasingly regarded as fundamental mechanisms for neural information processing and the adaptability and flexibility of neural systems. On the other hand, compared to traditional neural networks, memristive neural networks have more prominent advantages in applications such as chaotic encryption (Kong et al., 2024), associative memory (Wan et al., 2024), and pattern recognition (Deng et al., 2025a).

However, most of the existing studies of memristive neural networks have focused on continuous-time systems. In contrast to continuous systems that require solving complex differential equations, discrete-time models employ difference operations, which can significantly reduce computational complexity and storage requirements while still generating highly complex dynamic behavior through low-dimensional mathematical structures (Deng et al., 2025b; Gao et al., 2025c; Luo et al., 2025a). Therefore, the cryptanalysis based on discrete memristive chaotic systems is highly regarded in chaos-based encryption technologies, including efficient image and video encryption (Gao et al., 2025a,b). Existing studies have shown that DMNNs are a kind of bionic network with complex dynamics. They can more accurately simulate electrical activity and dynamic interaction between neurons, helping understand the generation of neurological diseases (Shang et al., 2023). Furthermore, DMNNs described by difference equations are compatible with the iterative architecture of digital circuits, making it easier to implement circuit design through FPGA and facilitating practical engineering applications (Deng et al., 2025e; Sun et al., 2025; Yu et al., 2024a). In recent years, researchers have made great progress in the dynamic analysis of discrete memristive neural networks. For example, Wei et al. (2025) designed a three-dimensional DMHNN based on discrete memristor simulating self-synapse, and analyzed multi-ring-like volumes and offset behaviors. Lu et al. (2024) proposed a discrete small-world neuronal network model based on the Rulkov neuron model and memristor synapse, and explored the influence of the neuron model parameter, network topology, and memristor coupling strength on the dynamic behavior of the network. Bao et al. (2025) coupled a memristor with an internal multi-segment state function into a two-neuron Hopfield neural network, constructing a novel discrete-time memristive two-neuron Hopfield neural network(DMT-HNN) capable of emerging complex multi-stripe/wave hyperchaos.

Currently, in the design of discrete chaotic systems based on DMNN, the artificial neural network models adopted by scholars are mainly various types of discrete neuronal networks, as well as discretized Hopfield neural networks and cellular neural networks. However, existing studies still lack sufficient exploration of discrete neural network architectures with novel topologies. It is worth noting that the bidirectional associative memory (BAM) neural network proposed by Kosko (2002) features a distinctive bidirectional feedback structure: neurons between different layers are fully interconnected, while no connections exist among neurons within the same layer. This network structure can not only support bidirectional hetero-association but also potentially provides an ideal model framework for the generation of nonlinear dynamics (Cao & Xiao, 2007; Huang et al., 2023; Song & Cao, 2007). To date, no research has introduced memristors into such artificial neural networks with bidirectional feedback structures to construct chaotic systems with rich dynamical behavior. In order to fill this research gap, this paper proposes a discrete bidirectional neural network (DBNN) based on the BAM neural network, and introduces a locally active discrete memristor as a synaptic connection into the DBNN model, resulting in a six-dimensional discrete bidirectional memristive neural network (DBMNN) model. Numerical simulations show that the DBMNN system has extremely complex dynamical behavior. Additionally, the digital circuit design of the proposed DBMNN is successfully implemented on a field-programmable gate array (FPGA).

The main contributions and innovative points of this article are summarized as follows:

(1)A neural network model featuring a novel topology, specifically a discrete bidirectional neural network (DBNN), is proposed. A locally active discrete memristor is introduced as a connection synapse into the DBNN, thereby constructing a discrete bidirectional memristive neural network (DBMNN) with complex dynamic characteristics.

(2)The chaotic behavior of the DBMNN is studied using multiple dynamic analysis methods. The analysis reveals that the system exhibits highly complex hyperchaos. Within specific parameter intervals, up to three or four positive Lyapunov exponents are observed. Transient behavior, heterogeneous coexisting attractors, and large-scale amplitude control are also found.

(3)The digital circuit design of the proposed DBMNN is implemented on an FPGA, with its dynamics verified through the combination of software simulation and hardware experiments.

The rest of the article is organized as follows. Section 2 presents a discrete locally active memristor and a general bidirectional discrete neural network model, and then constructs a specific DBMNN model based on them. Section 3 studies the dynamical behavior of the proposed DBMNN. Section 4 describes the FPGA experimental platform used to implement DBMNN. Finally, Section 5 provides a summary of the article.

2. Model description

In this section, we first propose a new discrete locally active memristor model, analyze its hysteresis loop and bistable behavior, and verify its non-volatility and locally active characteristic. Next, based on the structure of the BAM neural network, we present a discrete neural network model with a bidirectional feedback topology and give its general mathematical formulation. Finally, in order to obtain a chaotic system with excellent dynamical performance, we introduce the memristor into the original DBNN to replace a fixed connection between two neurons, thus arriving at the discrete bidirectional memristive neural network (DBMNN) model under study.

2.1. Discrete locally active memristor model

The locally active memristor can more effectively emulate synaptic connections, making it more attractive for constructing neural networks. Meanwhile, the local activity is considered a source of complexity, which can further enhance the chaotic characteristics of neural network systems. We make modifications to an existing mathematical model of a locally active discrete memristor (Lai & Yang, 2023) and obtain a new discrete locally active memristor model, which is defined as follows:

$$\begin{cases} i_n = W(\varphi_n)\, v_n, \\ W(\varphi_n) = \sin(\varphi_n), \\ \varphi_{n+1} = F(\varphi_n, v_n) = \alpha \sin(\varphi_n) + \beta v_n. \end{cases} \tag{1}$$

where φ_n is the state variable of the memristor and $W(\varphi_n)$ is memductance, i_n and v_n represent discrete input current and discrete output voltage, respectively. $F(\varphi_n, v_n)$ is the internal state function of the memristor, and parameters α and β influence the evolution of the internal state variable of the discrete memristor. To ensure that the discrete memristor satisfies the fundamental memristive properties while simultaneously possessing non-volatility and local activity, α and β are chosen as 1.1 and 0.1. For verification of the memristive characteristics of the model, the behavior of the discrete memristor under sinusoidal voltage signals $V_n = A\sin(\omega n)$ with different amplitudes and frequencies is investigated. By fixing a representative amplitude $A = 1$ to demonstrate

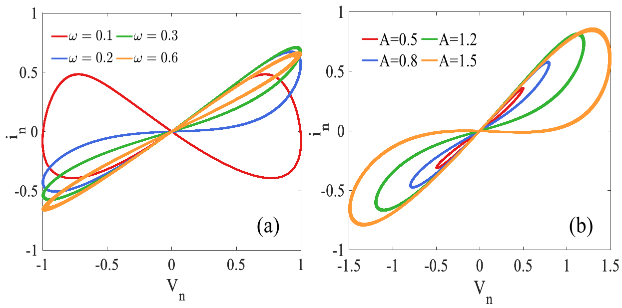

Fig. 1. The pinched hysteresis loops of the new locally active discrete memristor: (a) Fixed amplitude $A = 1$, and frequency ω for 0.1, 0.2, 0.3, 0.6; (b) Fixed frequency $\omega = 0.3$, and amplitude A for 0.5, 0.8, 1.2, 1.5.

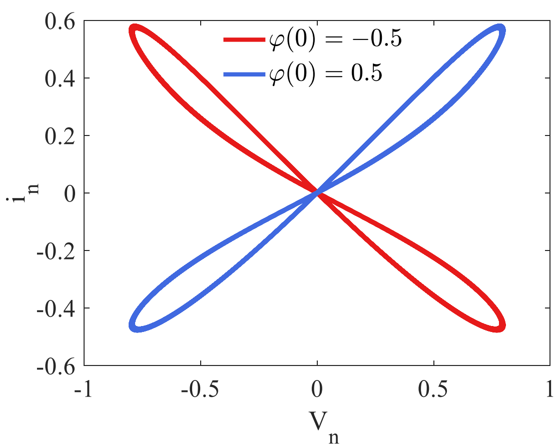

Fig. 2. Pinched hysteresis loops with different initial values $\varphi(0) = -0.5$ and $\varphi(0) = 0.5$.

the variation of the hysteresis loop with frequency clearly, and selecting the frequencies ω as 0.1, 0.2, 0.3, and 0.6, it is observed that the characteristic curves of the memristor are pinched hysteresis loops, and its area decreases with increasing input supply frequency, as depicted in Fig. 1(a). When the frequency $\omega = 0.3$ is fixed, Fig. 1(b) shows that the area of the hysteresis loop increases with the increase of amplitude A and its basic shape remains unchanged.

The memristor model can exhibit bistable behavior at appropriate values of amplitude and frequency. When $A = 0.8$ and $\omega = 0.3$, the memristor exhibits two distinct pinched hysteresis loops at initial values $\varphi(0) = -0.5$ and $\varphi(0) = 0.5$, as depicted in Fig. 2.

The non-volatility of a memristor refers to its ability to retain a resistance state to store information after the power is disconnected. The POP plot of a memristor can be used to verify its non-volatility. It can be derived from Eq. (1):

$$f\left(\varphi_n, v_n\right) = a \sin\left(\varphi_n\right) + \beta v_n - \varphi_{n+1} \tag{2}$$

Disconnect the input voltage of the memristor($v_n = 0$), then:

$$\varphi_{n+1} - \varphi_n = a \sin\left(\varphi_n\right) - \varphi_n \tag{3}$$

The curve of Eq. (3) is the POP diagram of the memristor, as shown in Fig. 3(a). The curve in the figure has two negative slopes intersecting the x-axis, indicating the existence of two stable equilibrium points for the memristor; therefore, the memristor is non-volatile.

The local activity of the memristor can be verified through a DC V-I diagram, assuming that $\varphi_n = \varepsilon$, $v_n = V$, $\varphi_{n+1} = \varphi_n$. Eq. (1) can be

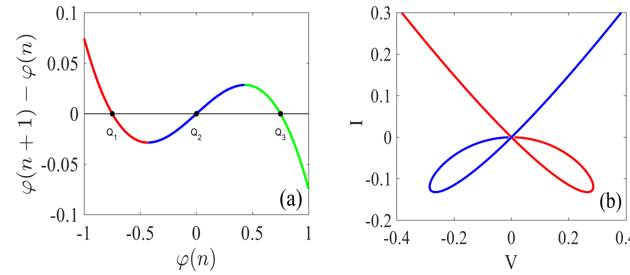

Fig. 3. Characteristic curves of local active memristor. (a) POP curve of the memristor. (b) DC V-I plot of memristor.

expressed as:

$$\begin{cases} V = \frac{\varepsilon}{\beta} - \frac{\alpha}{\beta}\sin(\varepsilon) \\ I = W(\varepsilon)V = \sin(\varepsilon)(\frac{\varepsilon}{\beta} - \frac{\alpha}{\beta}\sin(\varepsilon)) \end{cases} \tag{4}$$

where V and I represent the DC voltage and current, respectively. The DC V-I plot of the memristor is depicted in Fig. 3(b). The red region exhibits a negative slope, indicating local activity in the memristor.

2.2. Model of discrete bidirectional neural network

The traditional Bidirectional Associative Memory (BAM) neural network enables information to flow in both directions: the initial pattern can act on either layer of the network, and when an input signal is applied to one layer, the corresponding output can be obtained from the other layer. The weight matrix W of the BAM neural network describes the full connection between its two layers of neurons, while no feedback path exists among neurons within the same layer. The set of neuron state variables in two layers can iterate alternately in different dimensions, that is, when the network receives an m-dimensional vector input from the X layer, it may output an l-dimensional vector from the Y layer (Cao & Xiao, 2007; Huang et al., 2023; Kosko, 2002; Song & Cao, 2007). Based on the topological structure of BAM neural network, and considering the discrete model's advantages in computational resource and its high efficiency of generating chaotic systems, we propose a discrete bidirectional neural network (DBNN) model with asymmetric connections($W_{XY} \neq W_{YX}$) (Stamov, 2022), whose general mathematical definition is given as follows:

$$\begin{cases} x_i(n+1) = -px_i(n) + \sum_{j=1}^{m} c_{ji} f_j\left(y_j(n)\right) + I_i \\ y_j(n+1) = -qy_j(n) + \sum_{i=1}^{l} d_{ij} f_i\left(x_i(n)\right) + J_j, \end{cases} \tag{5}$$

where $x_i(n)$ and $y_j(n)$ respectively refer to the output state values of the ith neuron in the X layer and the jth neuron in the Y layer at the nth time instant, f_i, f_j represent the activation functions of neurons, c_{ji} is the weight of the feedback from neuron Y_j to neuron X_i, and d_{ij} is the weight of the feedback from neuron X_i to neuron Y_j. The terms $-px_i(n)$ and $-qy_j(n)$ describe the self-decay of the neuron output at the current time n, and p, q are constant parameters. I_i, I_j denote the external stimuli applied to the neurons. The topological structure of the general DBNN is shown in Fig. 4. The asymmetric weight connections between neurons are simplified as a single connection line for clarity.

2.3. Proposed discrete memristive bidirectional neural network

Inspired by the biological embodiment of the neural network structure with bidirectional connections in cortical circuits (Qiao et al., 2019; Ullman, 1995; Yagi et al., 1989), this article aims to explore the use of a simple bidirectional neural network model to simulate complex neurodynamic behavior. In this section, we design a specific DBNN model with three neurons (X_1, X_2, X_3) in the X layer and two neurons (Y_a, Y_b) in the Y layer, that is, $m = 3$ and $l = 2$. The two neurons in the Y layer are

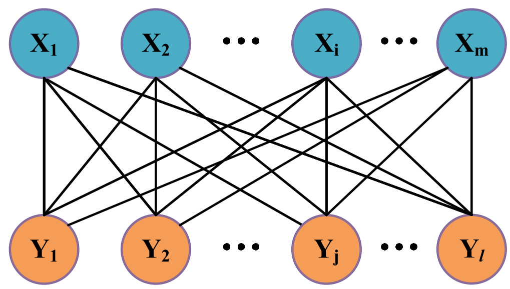

Fig. 4. The topological structure of the general DBNN model.

labeled as Y_a and Y_b (instead of Y_1 and Y_2) to avoid ambiguity in weight indexing between layers. We stipulate that all neurons uniformly use the hyperbolic tangent activation function. The self-decay constants of the all neurons outputs are set to 0.1, and the external stimuli applied to the neurons are set to zero. The feedback matrix W_{XY} from (X_1, X_2, X_3) to (Y_a, Y_b) and the feedback matrix W_{YX} from (Y_a, Y_b) to (X_1, X_2, X_3) determine the nonlinear characteristics of the system. Based on the principle of generating chaotic dynamics (Jin & Adamatzky, 2025), and in conjunction with extensive numerical experiments, the weight values are set as follows:

$$W_{XY} = \begin{bmatrix} -1 & 1.4 & -2.5 \\ w_{1b} & 1.5 & -4 \end{bmatrix}, \quad W_{YX} = \begin{bmatrix} w_{a1} & -2 \\ -3 & 4 \\ -0.4 & 2.5 \end{bmatrix} \tag{6}$$

According to Eqs. (5) and (6), the difference equation of the specific DBNN model can be derived as follows:

$$\begin{cases} x_1(n+1) = -0.1x_1(n) + w_{a1}\tanh(y_a(n)) - 2\tanh(y_b(n)) \\ x_2(n+1) = -0.1x_2(n) - 3\tanh(y_a(n)) + 4\tanh(y_b(n)) \\ x_3(n+1) = -0.1x_3(n) - 0.4\tanh(y_a(n)) + 2.5\tanh(y_b(n)) \\ y_a(n+1) = -0.1y_a(n) - \tanh(x_1(n)) + 1.4\tanh(x_2(n)) \\ \qquad -2.5\tanh(x_3(n+1)) \\ y_b(n+1) = -0.1y_b(n) + w_{1b}\tanh(x_1(n)) \\ \qquad +1.5\tanh(x_2(n)) - 4\tanh(x_3(n+1)). \end{cases} \tag{7}$$

To further investigate the effect of memristor introduction on the dynamics of the two-layer topological neural network, the proposed discrete locally active memristor is embedded into the feedback path of the DBNN in (7) by replacing the constant connection $w_{b3} = 2$ from neuron Y_b to neuron X_3. The resulting DBMNN model is shown in Fig. 5, and its difference equation is given as follows:

$$\begin{cases} x_1(n+1) = -0.1x_1(n) + w_{a1}\tanh(y_a(n)) - 2\tanh(y_b(n)) \\ x_2(n+1) = -0.1x_2(n) - 3\tanh(y_a(n)) + 4\tanh(y_b(n)) \\ x_3(n+1) = -0.1x_3(n) - 0.4\tanh(y_a(n)) \\ \qquad +kW(\varphi(n))\tanh(y_b(n)) \\ y_a(n+1) = -0.1y_a(n) - \tanh(x_1(n)) \\ \qquad +1.4\tanh(x_2(n)) - 2.5\tanh(x_3(n)) \\ y_b(n+1) = -0.1y_b(n) + w_{1b}\tanh(x_1(n)) \\ \qquad +1.5\tanh(x_2(n)) - 4\tanh(x_3(n)) \\ \varphi(n+1) = 1.1\sin(\varphi(n)) + 0.1\tanh(y_b(n)) \end{cases} \tag{8}$$

where $x_1(n)$, $x_2(n)$, $x_3(n)$, $y_a(n)$, and $y_b(n)$ represent the output state variables of each neuron at the nth iteration, φ is the internal state variable of the memristor, w_{a1} and w_{1b} are system parameters. The term $kW(\varphi(n))$ replaces the connection weight w_{b3}, where k is the coupling strength of the memristor and $W(\varphi(n)) = \sin(\varphi(n))$ is the memductance. The specified weight values serve as a basic configuration for the subsequent dynamical analysis, ensuring that the system's dynamics remain bounded while exhibiting a hyperchaotic state characterized by multiple positive Lyapunov exponents.

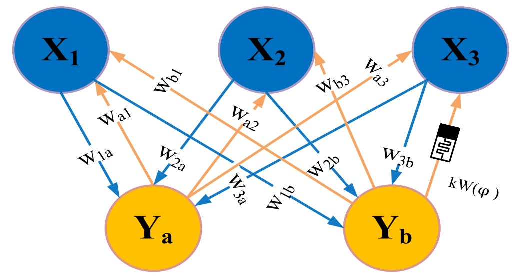

Fig. 5. Connection diagram of the proposed DBMNN model(letter indices used for clarity in weight notation).

Table 1
Equilibrium points and corresponding eigenvalues and stability.

Equilibrium points	Eigenvalues	Stability
$P_0(0,0,0,0,0,0)$	$(-1.9489 \pm 0.2618i, 1.7489 \pm 0.2618i, -0.1, 1.1)$	Unstable
$P_1(0,0,0,0,0,0.749)$	$(-1.1214 \pm 1.0478i, 0.9214 \pm 1.0478i, -0.1, 0.8056)$	Unstable
$P_2(0,0,0,0,0,-0.749)$	$(\pm 3.5413, -1.3909, -0.1, 1.1909, 0.8056)$	Unstable

3. Dynamics analysis of DBMNN

In this section, we comprehensively study the complex dynamical behavior of the proposed DBMNN using various numerical analysis methods, including equilibrium point analysis, phase diagrams, bifurcation diagrams, Lyapunov exponent spectrum, and bi-parameter dynamic maps. The numerical analyses involving iterations in this paper are performed on a discrete-time dynamical system described by difference equations with a step size of $n = 1$, an iteration scale of N = 20,000 steps, and a transient length of 2000 steps.

3.1. Equilibrium point analysis

Equilibrium point analysis is essential for understanding the overall complexity of nonlinear systems and elucidating the evolution of chaos. The equilibrium points and stability of the DBMNN are determined using graphical and numerical methods. By setting $x_i(n+1) = x_i(n)$, $y_j(n+1) = y_j(n)$, and $\varphi(n+1) = \varphi(n)$ to make the next state variable equal to the current state variable, the following equilibrium point equation can be derived:

$$\begin{cases} x_1^* = -0.1x_1^* + w_{a1}\tanh(y_a^*) - 2\tanh(y_b^*) \\ x_2^* = -0.1x_2^* - 3\tanh(y_a^*) + 4\tanh(y_b^*) \\ x_3^* = -0.1x_3^* - 0.4\tanh(y_a^*) \\ \qquad +kW(\varphi)\tanh(y_b^*) \\ y_a^* = -0.1y_a^* - \tanh(x_1^*) \\ \qquad +1.4\tanh(x_2^*) - 2.5\tanh(x_3^*) \\ y_b^* = -0.1y_b^* + w_{1b}\tanh(x_1^*) \\ \qquad +1.5\tanh(x_2^*) - 4\tanh(x_3^*) \\ \varphi^* = 1.1\sin(\varphi^*) + 0.1\tanh(y_b^*) \end{cases} \tag{9}$$

From the sixth equation in Eq. (9), we can obtain:

$$\tanh(y_b^*) = \left[\varphi^* - 1.1\sin(\varphi^*)\right]/0.1 \tag{10}$$

By substituting Eq. (10) into the first three equations of Eq. (9) and simplifying them, the state variables x_1^*, x_2^*, and x_3^* can be represented by

$$\begin{cases} x_1^* = \left[w_{a1}\tanh(y_a^*) - 20\varphi^* + 22\sin(\varphi^*)\right]/1.1 \\ x_2^* = \left[-3\tanh(y_a^*) + 40\varphi^* - 44\sin(\varphi^*)\right]/1.1 \\ x_3^* = \left[-0.4\tanh(y_a^*) + 10kW(\varphi^*)\varphi^* \right. \\ \qquad \left. -11kW(\varphi^*)\sin(\varphi^*)\right]/1.1 \end{cases} \tag{11}$$

To reveal the stability of equilibrium points when the system operates in typical hyperchaotic states, and to provide a consistent reference

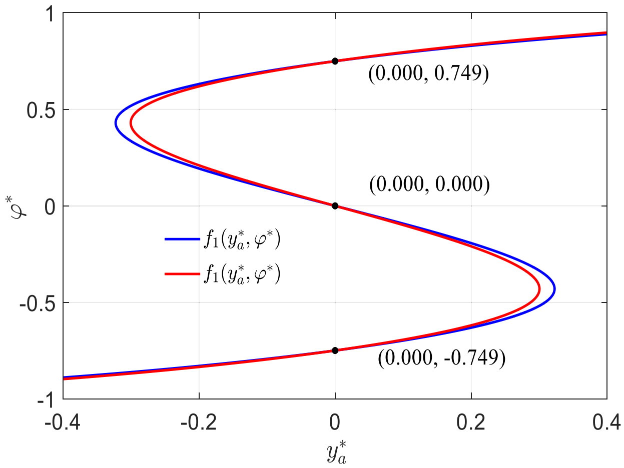

Fig. 6. The curves of the functions $f_1(y_a^*, \varphi^*)$, $f_1(y_a^*, \varphi^*)$, and their intersection points with $w_{a1} = 2.5, w_{1b} = -3.2, k = 2.5$.

point for the subsequent investigation of the influence of parameters on dynamics, we set w_{a1}, w_{1b}, and k to 2.5, -3.2, and 2.5. Substitute Eq. (11) into the fourth and fifth equations of Eq. (9) and replace $\tanh(y_b^*)$ with $10\varphi^* - 11\sin(\varphi^*)$ to eliminate y_b^*. Finally, two curve functions involving only y_a^* and φ^* are obtained, which are expressed as

$$\begin{cases} f_1(y_a^*, \varphi^*) & = 1.1y_a^* + \tanh\left(r_1(y_a^*, \varphi^*)\right) \\ & -1.4\tanh\left(r_2(y_a^*, \varphi^*)\right) \\ & +2.5\tanh\left(r_3(y_a^*, \varphi^*)\right) \\ f_2(y_a^*, \varphi^*) & = \tanh\left\{\left(-3.2\tanh\left(r_1(y_a^*, \varphi^*)\right)\right.\right. \\ & +1.5\tanh\left(r_2(y_a^*, \varphi^*)\right) \\ & \left.\left.-4\tanh\left(r_3(y_a^*, \varphi^*)\right)\right)/1.1\right\} \\ & -10\varphi^* + 11\sin(\varphi^*) \end{cases} \tag{12}$$

where

$$\begin{cases} r_1(y_a^*, \varphi^*) & = \left[2.5\tanh(y_a^*) - 20\varphi^* + 22\sin(\varphi^*)\right]/1.1 \\ r_2(y_a^*, \varphi^*) & = \left[-3\tanh(y_a^*) + 40\varphi^* - 44\sin(\varphi^*)\right]/1.1 \\ r_3(y_a^*, \varphi^*) & = \left[-0.4\tanh(y_a^*) + 25W(\varphi^*)\varphi^*\right. \\ & \left.-27.5W(\varphi^*)\sin(\varphi^*)\right]/1.1 \end{cases} \tag{13}$$

The two curves described by Eq. (12) are plotted in different colors in Fig. 6, with their intersection points representing the system's equilibrium points. The system exhibits three equilibrium points: $P_0(0,0,0,0,0,0)$, $P_1(0,0,0,0,0,0.749)$, and $P_2(0,0,0,0,0,-0.749)$. Their stability is determined by analyzing the eigenvalue properties of the Jacobian matrix, which is given as follows:

$$J = \begin{bmatrix} -0.1 & 0 & 0 & 2.5e_1 & -2e_2 & 0 \\ 0 & -0.1 & 0 & -3e_1 & 4e_2 & 0 \\ 0 & 0 & -0.1 & -0.4e_1 & 2.5h_1 & 2.5h_2 \\ -d_1 & 1.4d_2 & -2.5d_3 & -0.1 & 0 & 0 \\ -3.2d_1 & 1.5d_2 & -4d_3 & 0 & -0.1 & 0 \\ 0 & 0 & 0 & 0 & 0.1h_1 & 1.1\cos(\varphi) \end{bmatrix} \tag{14}$$

where $d_1 = \mathrm{sech}^2(x_1)$, $d_2 = \mathrm{sech}^2(x_2)$, $d_3 = \mathrm{sech}^2(x_3)$, $e_1 = \mathrm{sech}^2(y_a)$, $e_2 = \mathrm{sech}^2(y_b)$, $m = \mathrm{sech}^2(\varphi)$, $h_1 = \sin(\varphi)e_2$, $h_2 = \cos(\varphi)\tanh(y_b)$.

Substitute the equilibrium point $P_1(x_1, x_2, x_3, y_a, y_b, \varphi) = (0,0,0,0,0,0.749)$ into the Jacobian matrix Eq. (14). By solving $|J - \lambda E| = 0$, we can obtain its corresponding eigenvalues: $\lambda_{1,2} = -1.1214 \pm 1.0478i$, $\lambda_{3,4} = 0.9214 \pm 1.0478i$, $\lambda_5 = -0.1000$, $\lambda_6 = 0.8056$. An equilibrium point is stable if the modulus of all its eigenvalues is less than 1; If any absolute eigenvalue is greater than 1, the equilibrium point is unstable. Obviously, the equilibrium point P_1 is unstable. The other two equilibrium points, P_0 and P_2, are both unstable by calculating their eigenvalues. Table 1 presents the three equilibrium points and their corresponding eigenvalues and stability.

3.2. Effect of coupling strength k on dynamics

To analyze the influence of memristor coupling strength, the parameters w_{a1} and w_{1b} are fixed at 2.5 and -3.2, with initial values set to $(0.1, 0.1, 0.1, 0.1, 0.1, 0.1)$. The Lyapunov exponent diagrams and bifurcation diagrams of DBMNN with the variation of parameter k are shown in Fig. 7 (Only the first four Lyapunov exponents are plotted in Fig. 7(a) for the convenience of display and analysis). The Lyapunov exponents are computed using the standard QR decomposition method for discrete maps (Bremen et al., 1997). At each iteration, a QR factorization is performed on the product of the Jacobian and the previous orthogonal matrix Q, and the logarithms of the absolute values of the diagonal elements in the resulting upper triangular matrix R are accumulated and averaged to obtain the Lyapunov exponents. The bifurcation diagrams are plotted by discarding the first 85% of iterations as transient and recording the remaining iterative values of state variables over the corresponding varying parameter. The selected parameters are varied with a step size of 0.005. It is known that a system exhibits chaotic behavior when its maximum Lyapunov exponent is positive, and the presence of two or more positive Lyapunov exponents indicates hyperchaos. As shown in Fig. 7(a) and (b), the DBMNN system demonstrates highly complex dynamics, existing in a hyperchaotic state over the majority of the parameter range. The presence of wide hyperchaotic intervals indicates that the system can robustly maintain complex dynamics under continuous parameter variations, which is desirable for applications requiring high complexity and extensive chaotic regions, as well as reflecting enhanced chaotic performance induced by the memristor and structural robustness of the proposed model.

The DBMNN system exhibits chaotic behavior when $k = 1.34$ and it initially resides in a hyperchaotic state characterized by two positive Lyapunov exponents. After a brief periodic window around $k = 1.5$, the system enters a longer parameter interval $k \in [1.5, 2.8]$ in which stable hyperchaotic behavior is sustained. For this hyperchaotic interval, a third positive Lyapunov exponent emerges when $k > 1.97$. Within $k \in [2.3, 2.6]$, a fourth positive Lyapunov exponent emerges, and the third exponent increases beyond 0.1, indicating that the system enters a high-dimensional hyperchaotic state characterized by exponential divergence of minor perturbations along four independent directions, marked by enhanced sensitivity to initial conditions and significantly increased unpredictability. As k continues to increase, after $k = 2.83$, the system transitions from hyperchaos to alternating weak chaos and periodic behavior. The bifurcation diagram and Lyapunov exponent diagram exhibit consistent dynamic behavior. In addition, from the bifurcation diagram, it is observed that within $k \in [1.5, 2.5]$, although the system's chaotic properties continuously intensify, the amplitude range of the state variable y_b gradually contracts.

Phase diagrams provide an intuitive visualization of the system's dynamical trajectories, revealing the nonlinear relationships between state variables and illustrating how system parameters influence the morphology of attractors. We plot the phase diagrams of x_2 and y_b for DBMNN with different k as shown in Fig. 8. At $k = 1.4$, Fig. 8(a) presents a hyperchaotic attractor with two positive Lyapunov exponents. At $k = 1.51$, Fig. 8(b) shows the attractor in the periodic window. When k increases to 2.5, Fig. 8(c) shows a hyperchaotic attractor with four positive Lyapunov exponents, compared to Fig. 8(a), the amplitude ranges of the state variables x_2 and y_b are reduced, while the ergodicity of trajectory points in the phase plane is enhanced. At $k = 2.92$, Fig. 8(d) presents an attractor exhibiting weak chaos. The properties of these attractors at different k values are consistent with the dynamical behaviors described in Fig. 7.

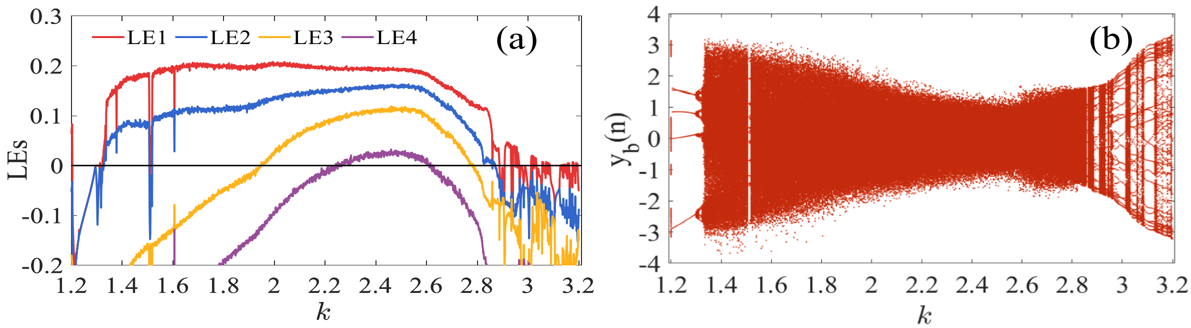

Fig. 7. Lyapunov exponent and bifurcation diagrams of the DBMNN with $k \in [1.25, 3.2]$, initial values for (0.1, 0.1, 0.1, 0.1, 0.1, 0.1), $w_{a1} = 2.5$, $w_{1b} = -3.2$. (a) Lyapunov exponent diagram. (b) Bifurcation diagram.

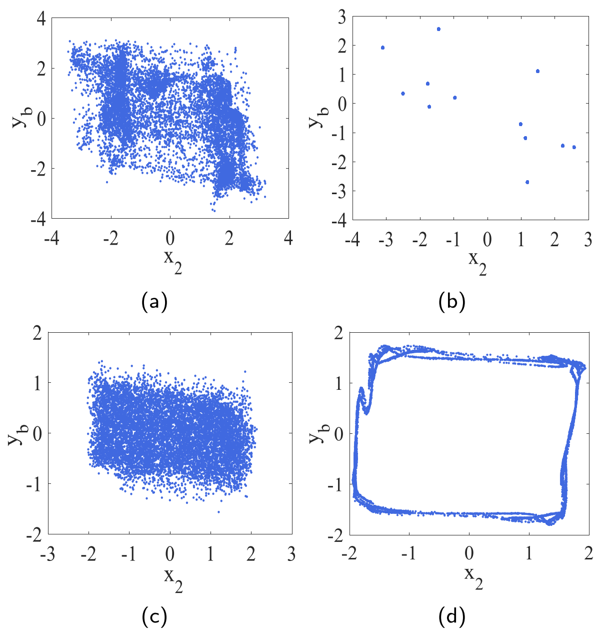

Fig. 8. Variation of phase diagrams of DBMNN under different k. (a) $k = 1.4$. (b) $k = 1.51$. (c) $k = 2.5$. (d) $k = 2.92$.

3.3. Effect of feedback weight w_{a1} and w_{1b} on dynamics

First, we study the effect of feedback weight w_{a1} on the dynamical behavior of DBMNN, with fixed parameters $k = 2.5, w_{1b} = -3.2$, and initial values set to (0.1, 0.1, 0.1, 0.1, 0.1, 0.1). Fig. 9 shows the Lyapunov exponent diagram and the bifurcation diagram of the DBMNN under the variation of parameter w_{a1} from 1.8 to 3.4. As can be seen from Fig. 9(a) and (b), the system has a considerable number of hyperchaotic intervals and periodic windows. In both intervals $[1.9, 2.32]$ and $[2.76, 3.25]$, there exists a phenomenon of alternating chaotic and periodic behavior. The difference is that the chaos in the former interval belongs to the category of hyperchaos, while in the latter interval it is ordinary weak chaos. The numerous periodic windows embedded in chaotic and hyperchaotic regions indicate a rich bifurcation structure of the DBMNN with respect to w_{a1}, and fine parameter variations can induce transitions between ordered and disordered dynamics. From a biological perspective, such transitions correspond to changes in neuronal activity patterns, where a shift from chaotic to periodic behavior may be abnormal, potentially offering insights into certain neurological conditions.

Moreover, it is evident from the bifurcation diagram that the amplitude range of the state variable y_b increases with increasing w_{a1}, and after $w_{a1} = 2.76$(where hyperchaos disappears), the amplitude range of y_b suddenly expands by approximately double. The DBMNN system exhibits continuous hyperchaotic behavior that is not interrupted by periodic behavior within $w_{a1} \in [2.32, 2.76]$, and further narrowing the interval to $[2.4, 2.6]$, a fourth positive Lyapunov exponent appears in Fig. 9(a).

We select another pair of state variables y_b and φ to plot their phase diagrams, as depicted in Fig. 10. When $w_{a1} = 1.99$, the dynamical behavior of the system is periodic. When $w_{a1} = 2.1$, the system exhibits hyperchaotic dynamics. At $w_{a1} = 2.45$, Fig. 10(c) shows a hyperchaotic attractor characterized by four positive Lyapunov exponents, with a larger region and stronger ergodicity than that depicted in Fig. 10(b). Continuing to increase w_{a1} to 3, Fig. 10(d) presents a weaker chaotic attractor, whose range expands but its ergodicity in the phase plane weakens, with trajectories predominantly distributed along the periphery.

For the influence of w_{1b} on dynamics, we fix the parameters $k = 2.5, w_{a1} = 2.5$, and set the initial values to (0.1, 0.1, 0.1, 0.1, 0.1, 0.1). The Lyapunov exponent diagram and the bifurcation diagram of DBMNN under the variation of parameter w_{1b} from -3.55 to -1.8 are shown in Fig. 11. From Fig. 11(a) and (b), it can be seen that the DBMNN system demonstrates hyperchaotic behavior within the parameter interval $w_{1b} \in [-3.3, -2.6]$, where the subinterval $[-3.3, -3.1]$ has four positive Lyapunov exponents, and $[-3.1, -2.76]$ contains three positive Lyapunov exponents. In addition, the system exhibits complex bifurcation phenomena under parameter variations of $w_{1b} \in [-2.6, -1.8]$. As w_{1b} increases, its dynamics transition from a chaotic state to a periodic state, then undergo period-doubling processes and re-enter a weak chaotic state through periodic windows. This cycle of chaos-period-chaos repeats multiple times, ultimately forming a complex period-doubling bifurcation region that is distinct from the dynamical pattern of the hyperchaotic region.

Similarly, we draw the phase diagrams of DBMNN as shown in Fig. 12. Fig. 12(a), (b), and (c) demonstrate the hyperchaotic attractors formed by the system under different w_{1b}, with the numbers of positive Lyapunov exponents being 3, 4, and 2, respectively. Fig. 12(d) shows the chaotic attractor in the period-doubling bifurcation region.

3.4. Bi-parameter based dynamical analysis

Although single-parameter analysis can reveal the bifurcation and chaos evolution of nonlinear systems along a specific path, it fails to characterize how multiple parameters jointly regulate dynamical behavior. To further investigate the influence of bi-parameter on the system's chaotic dynamics—particularly to identify specific regions capable of generating hyperchaotic behavior, we conduct a bi-parameter co-scan analysis of the DBMNN system based on the previous Lyapunov exponential analysis. Selecting the parameter varying regions as $k \in [1.8, 3]$, $w_{a1} \in [2, 3.2]$ and $w_{1b} \in [-3.5, -2]$, initial values for (1,1,1,1,1,1), the

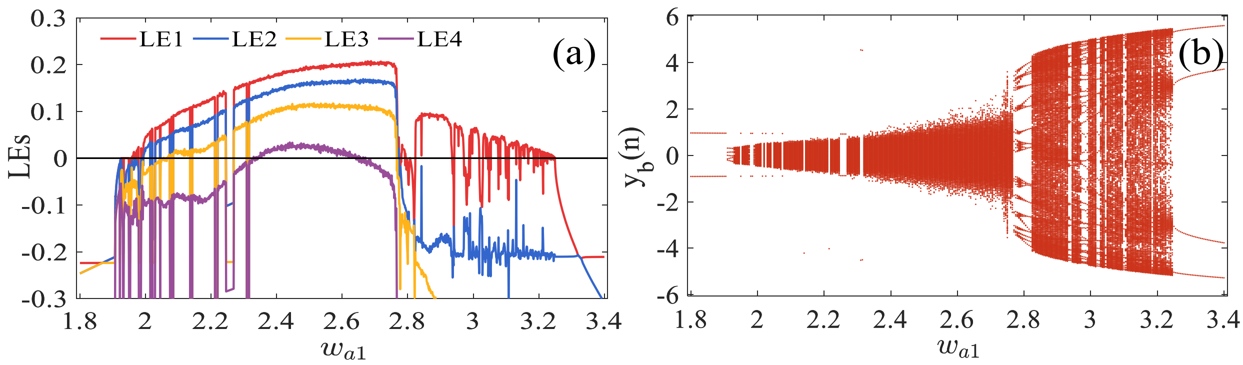

Fig. 9. Lyapunov exponent and bifurcation diagrams of DBMNN with $w_{a1} \in [1.8, 3.4]$, initial values for (0.1, 0.1, 0.1, 0.1, 0.1, 0.1), $k = 2.5, w_{1b} = -3.2$. (a) Lyapunov exponent diagram. (b) Bifurcation diagram.

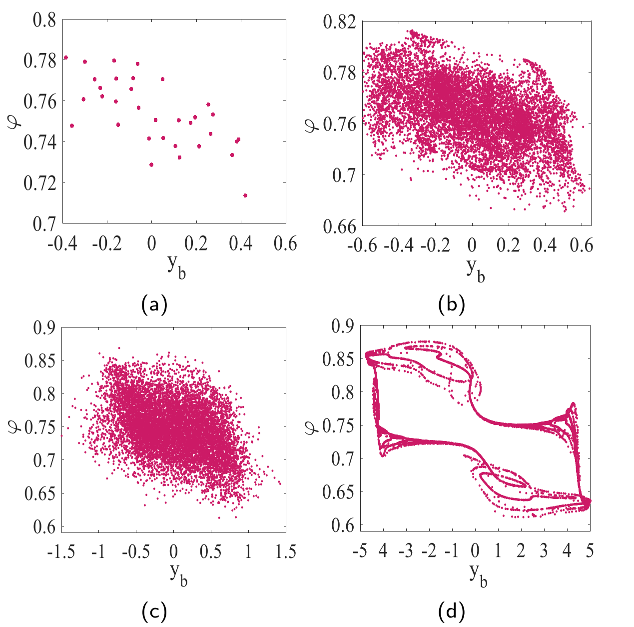

Fig. 10. Variation of phase diagrams of DBMNN under different w_{a1}. (a) $w_{a1} = 1.99$. (b) $w_{a1} = 2.1$. (c) $w_{a1} = 2.45$. (d) $w_{a1} = 3$.

bi-parameter dynamic maps of DBMNN are obtained by respectively fixing $w_{1b} = -3.2$,$w_{a1} = 2.5$ and $k = 2.5$, as shown in Fig. 13(a) to (c).

Colors in Fig. 13 are assigned based on the number of positive Lyapunov exponents (LEs) and the magnitude of the largest Lyapunov exponent at each parameter coordinate. Red corresponds to a hyperchaotic state with four positive LEs, magenta corresponds to hyperchaos with three positive LEs, orange denotes hyperchaos with two positive LEs, and yellow represents ordinary chaos. Green indicates a quasi-periodic state between chaos and periodicity, where the system exhibits very weak chaotic characteristics. Cyan corresponds to periodic state, and blue signifies stable equilibrium points. As shown in Fig. 13, the proposed system has extensive hyperchaotic regions in the two-parameter plane, which is consistent with the continuous hyperchaotic intervals reflected in previous single-parameter analyses. Therefore, we can achieve more comprehensive and efficient switching control over the system's dynamical modes from the perspective of multidimensional parameter regulation.

3.5. Heterogeneous coexisting attractors

The proposed DBMNN can exhibit heterogeneous coexisting attractors under different initial conditions of the memristor. Specifically, by selecting four groups of (w_{2b}, w_{3b}) while keeping the other weight parameters fixed(including $k = 2.5$, $w_{a1} = 2.5$, $w_{1b} = -3.2$), and the initial states of the neurons are set to $(0.1, 0.1, 0.1, 0.1, 0.1)$. Fig. 14 illustrates four types of coexisting asymmetric attractors observed under the above configurations, namely hyperchaotic(four positive LEs)-periodic, periodic-hyperchaotic(two positive LEs), hyperchaotic(two positive LEs)-weak chaotic(LE1 = 0.094, LE2 = −0.219), and quasi-periodic(LE1 = 0)-chaotic(LE1 = 0.134, LE2 = −0.101). In each case, the blue attractors correspond to the memristive initial condition $\varphi(0) = 0.5$, while the red attractors are generated with $\varphi(0) = -0.5$.

3.6. Transient chaos and chaotic attractor jumping

Transient chaos is a special transient behavior on the time scale. It refers to a dynamical phenomenon in which a system exhibits chaotic behavior within a finite time under certain parameter conditions, then transitions into a periodic or stable state. Set parameters $w_{a1} = 2.5$, $w_{1b} = -3.3$ and $k = 2$, and let the initial values be (1,1,1,1,1,0.1). The proposed DBMNN can exhibit typical transient chaos. As shown in Fig. 15(a), the dynamical behavior of the system transitions from a chaotic state to a periodic state after approximately 8900 iterations. Fig. 15(b) and (c) show the system's attractors in the chaotic and periodic regions, respectively.

Chaotic attractor jumping is also a kind of transient behavior. Unlike transient chaos, the system does not transition to a periodic state after a finite number of iterations, but rather enters another chaotic state, as evidenced by changes in both the time series and attractors. Set parameters $w_{a1} = 2.6$, $w_{1b} = -2.8$, $w_{2a} = 1.5$ and $k = 1.86$, and let the initial values also be (1,1,1,1,1,0.1). The system undergoes a chaotic attractor jumping after approximately 6700 iterations, manifested as significant changes in the waveform characteristics and amplitude range of the x_1 time series in Fig. 16(a), as well as the two distinct shapes of chaotic attractors presented in Fig. 16(b) and (c). In transient behavior, the transition from chaos to periodicity is irreversible, while different aperiodic states can be transformed into each other as shown in Fig. 17.

3.7. Large-scale amplitude control by $wb2$

Large-scale amplitude control refers to the ability of a nonlinear system to adjust and expand the oscillation amplitude of its state variables over a wide range of a parameter without disrupting its original dynamic state, which is essential for maintaining dynamical stability and preventing signal divergence.

Set $w_{a1} = 2.5$, $w_{1b} = -3.2$ and $k = 2.5$. And let the initial values be (1, 1, 1, 1, 1, 1). The system can exhibit a wide range of amplitude control determined by the parameter w_{b2}. As shown in Fig. 18, as w_{b2} increases significantly to 200, 400, 600, and 1000, the attractors of the system remain chaotic, while their amplitude in the x_2 dimension expands to the corresponding scale.

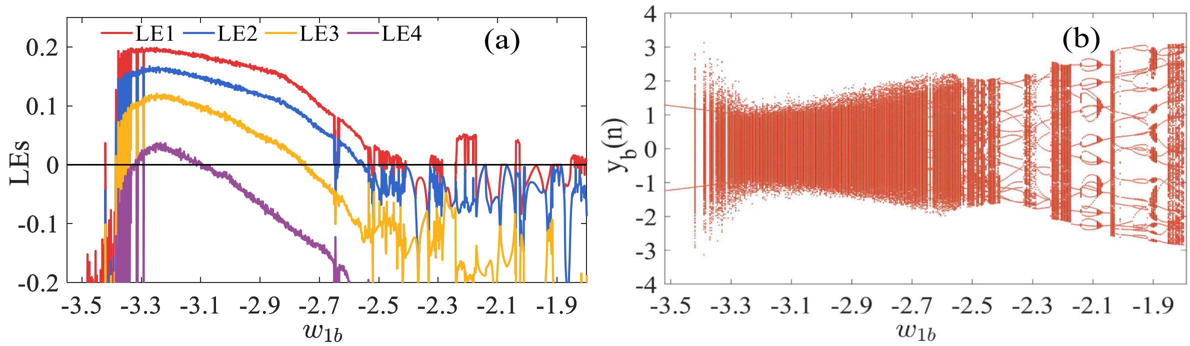

Fig. 11. Lyapunov exponent and bifurcation diagrams of DBMNN with $w_{1b} \in [-3.55, -1.8]$, initial values for (0.1, 0.1, 0.1, 0.1, 0.1, 0.1), k= 2.5, $w_{a1} = 2.5$. (a) Lyapunov exponent diagram. (b) Bifurcation diagram.

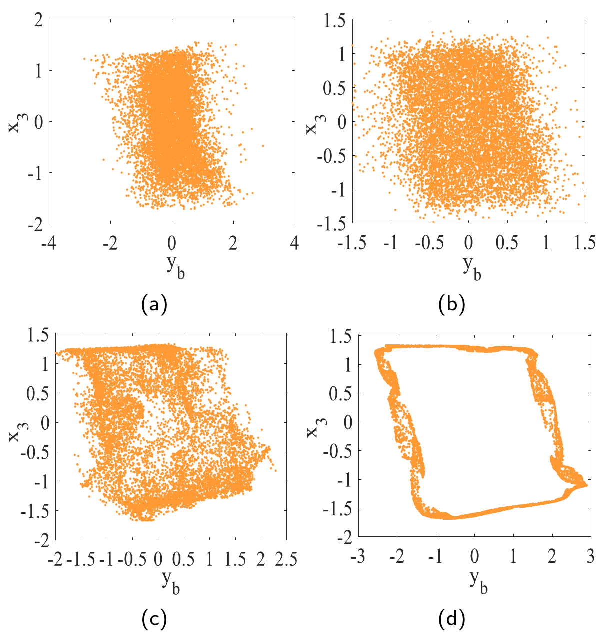

Fig. 12. Variation of phase diagrams of DBMNN under different w_{1b}. (a) $w_{1b} = -3.36$. (b) $w_{1b} = -3.25$. (c) $w_{1b} = -2.6$. (d) $w_{1b} = -2.04$.

3.8. Sample entropy of the DBMNN

Sample entropy(SampEn) is an effective method for quantitatively analyzing the complexity or unpredictability of time series, which exhibits improved consistency and reduced dependence on data length compared to approximate entropy. In chaotic systems, a larger sample entropy value indicates a higher degree of complexity and stronger randomness in the generated sequence. Such characteristics are greatly desirable in practical applications, including secure communications, chaotic encryption, and pseudo-random sequence generation. The sample entropy of a signal sequence x(n) is defined as Richman and Moorman (2000):

$$\text{SampEn}(m, r) = -\ln\left(\frac{C(m+1, r)}{C(m, r)}\right) \tag{15}$$

where $C(m, r)$ denotes the number of pairs of m-dimensional vectors whose Chebyshev distance is less than r. Setting the parameters $w_{a1} = 2.5$, $w_{1b} = -3.2$ and the initial conditions as $(0.1, 0.1, 0.1, 0.1, 0.1, 0.1)$, the SampEn of DBMNN is calculated based on the state sequences $x_2(n)$, $y_b(n)$, and $\varphi(n)$, respectively. The variation of SampEn with respect to k is shown in Fig. 19. The SampEn of DBMNN increases within the range of $k \in [1.5, 2.6]$ (consistent with the appearance of the third/fourth positive Lyapunov exponent in Fig. 7a), and reaches a maximum of 2.0835. This indicates that the proposed DBMNN can generate high-quality chaotic sequences with significant potential for encryption applications.

4. FPGA-based circuit implementation of DBMNN

Circuit implementation is a critical step in verifying the dynamical characteristics of chaotic systems and promoting their engineering applications. Such validation is essential for confirming the presence of chaotic behavior and assessing the robustness and accuracy of theoretical models in the actual circuit environment (Tian et al., 2025b). Currently, there are mainly two methods for circuit design. One is to use discrete circuit components (resistors, capacitors, inductors, operational amplifiers, etc.) to build traditional hardware circuits. This method has difficulties in error correction and parameter adjustment. Another method is to use a Field Programmable Gate Array (FPGA) for digital circuit design. FPGA-based circuit implementations offer numerous advantages, including flexible parameter tuning, high stability, fast computational speed, and accurate iterative computation.

In this paper, the circuit design of the proposed DBMNN model is implemented on an FPGA platform. Specifically, we implement the DBMNN model based on Verilog HDL hardware design language and IEEE 754–1985 high precision 32-bit floating point standard on the Vivado 2018.3 design platform using the Xilinx FPGA development board ZYNQ-XC7Z020-CLG400-1. Notably, the activation functions employed by neurons in the DBMNN are hyperbolic tangent functions. However, this activation function with exponential forms is not suitable for FPGA-based digital implementation, due to the limitations in hardware resources and power consumption. To address this issue, Kwan proposed a simple second-order piecewise nonlinear function that exhibits a tanh-like transition between the lower and upper saturation regions (Kwan, 1992). This function is not only well-suited for direct hardware implementation but also effectively simulates the numerical characteristics of the hyperbolic tangent function. Its expression is given as follows:

$$G(x) = \begin{cases} 1 & x \geq L \\ H_s(x) & -L < x < L \\ -1 & x \leq -L \end{cases} \tag{16}$$

where

$$H_s(x) = \begin{cases} x(r - \theta x), & 0 \leq x < L \\ x(r + \theta x), & -L < x < 0 \end{cases} \tag{17}$$

where r and θ represent the slope and gain of $H_s(x)$ between $-L < x < L$. By imposing the following conditions: $H_s(L) = 1$, $H_s(-L) = -1$, $H_s'(L^-) = H_s'(-L^+) = 0$, and $H_s'(0) = \tanh'(0) = 1$, the parameters r, θ, and L can be determined as 1, 0.25, and 2, respectively. Fig. 20 shows the curves of the hyperbolic tangent function and the second-order piecewise function for visual comparison. The root mean squared relative

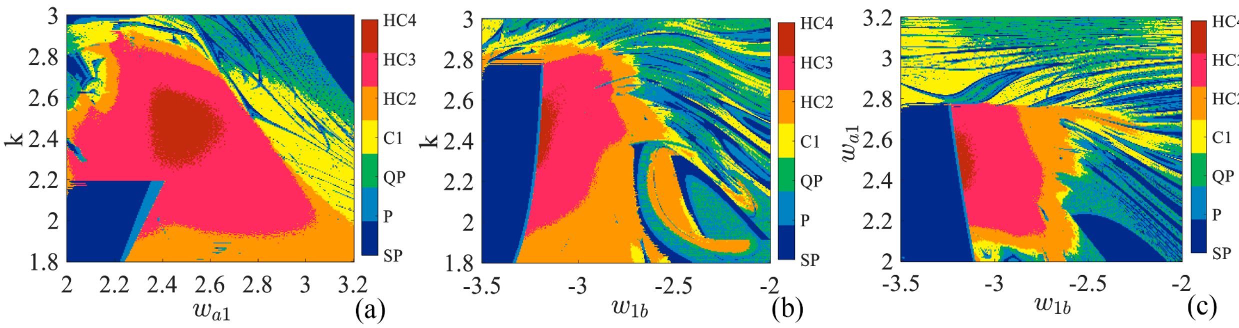

Fig. 13. Bi-parameter dynamic maps: (a)$w_{a1} - k$ plane with $w_{1b} = -3.2$. (b)$w_{1b} - k$ plane with $w_{a1} = 2.5$. (c)$w_{1b} - w_{a1}$ plane with $k = 2.5$.

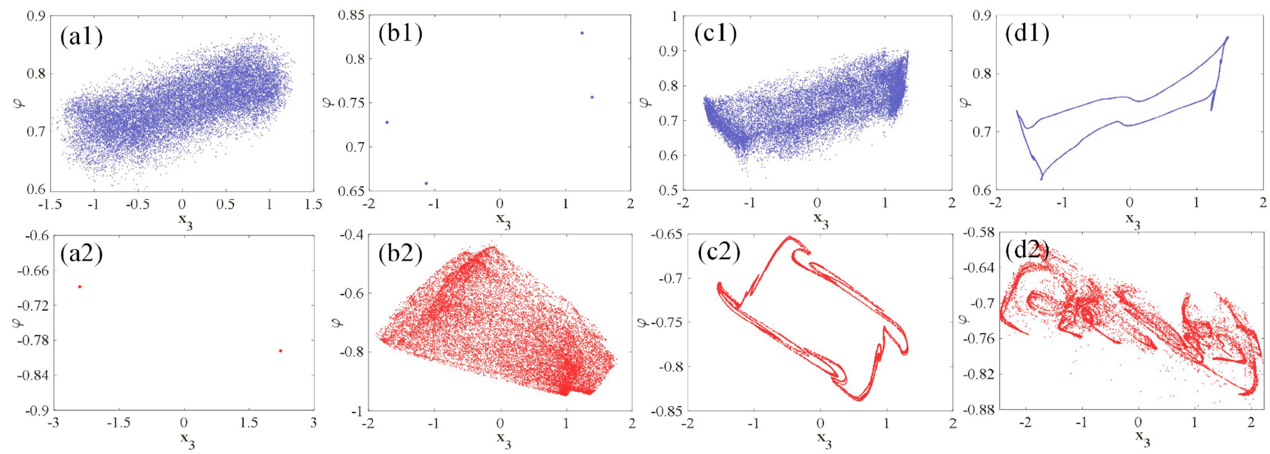

Fig. 14. The heterogeneous coexisting attractors of DBMNN with parameters $w_{a1} = 2.5, k = 2.5, w_{1b} = -3.2$, and initial values $(0.1, 0.1, 0.1, 0.1, 0.1, 0.5)$(in blue) and $(0.1, 0.1, 0.1, 0.1, 0.1, -0.5)$(in red): (a1)(a2)Hyperchaotic and periodic coexisting attractors with $(w_{2b}, w_{3b}) = (1.5, -4)$; (b1)(b2)Periodic and hyperchaotic coexisting attractors with $(w_{2b}, w_{3b}) = (-3.3, -1)$; (c1)(c2)Hyperchaotic and weak chaotic coexisting attractors with $(w_{2b}, w_{3b}) = (-3.25, 1.3)$; (d1)(d2)Quasi-periodic and chaotic coexisting attractors with $(w_{2b}, w_{3b}) = (-1.08, -2.9)$.

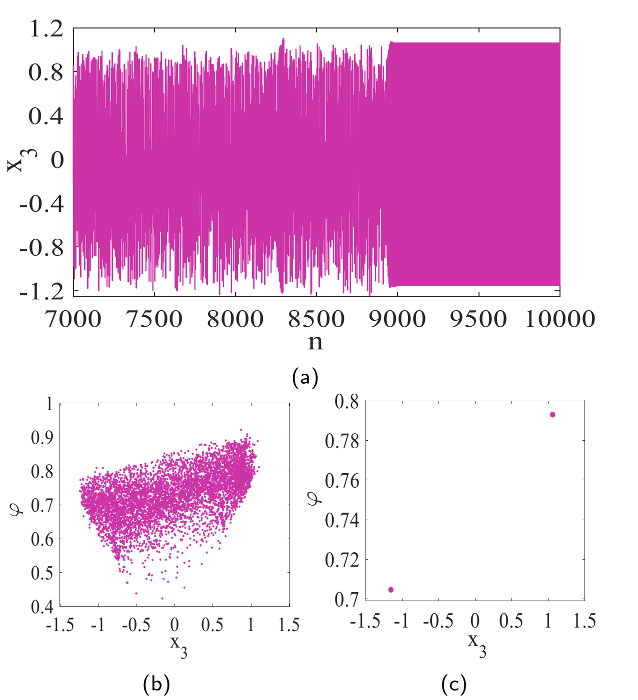

Fig. 15. Transient chaotic time series and phase diagrams of DBMNN with $w_{a1} = 2.5, w_{1b} = -3.3$, and $k = 2$: (a) Time series of x_3. (b) Phase diagram of the chaotic region. (c)Phase diagram of the periodic region.

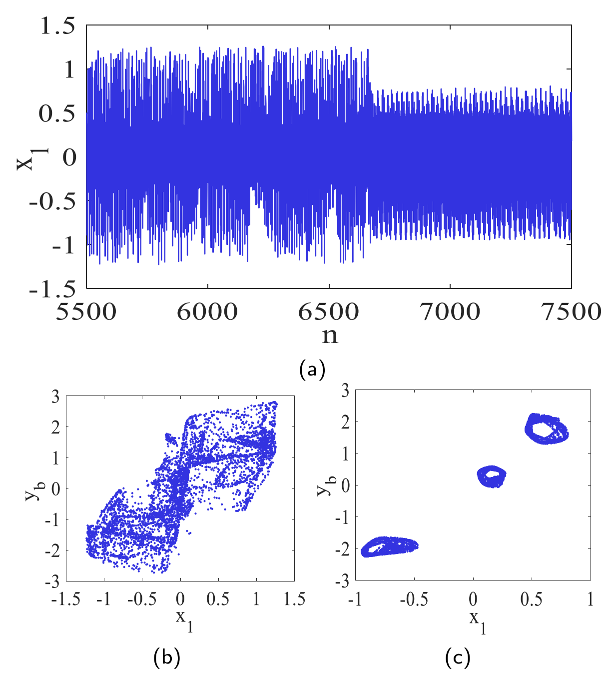

Fig. 16. Chaotic attractor jumping: time series and phase diagrams of DBMNN with $w_{a1} = 2.6, w_{1b} = -2.5, w_{2a} = 1.5$, and $k = 1.86$: (a) Time series of x_1. (b) Phase diagram of the first chaotic region. (c)Phase diagram of the second chaotic region.

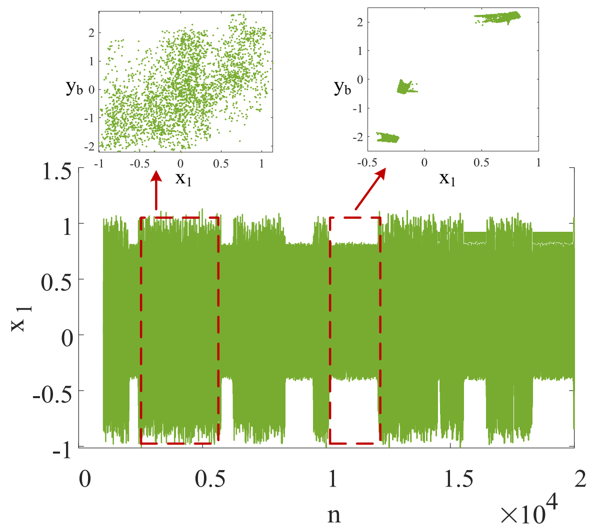

Fig. 17. Random-like jumps between two distinct attractor states with $w_{a1} = 2.62$, $w_{1b} = -2.75$, $w_{2a} = 1.4$, and $k = 1.86$.

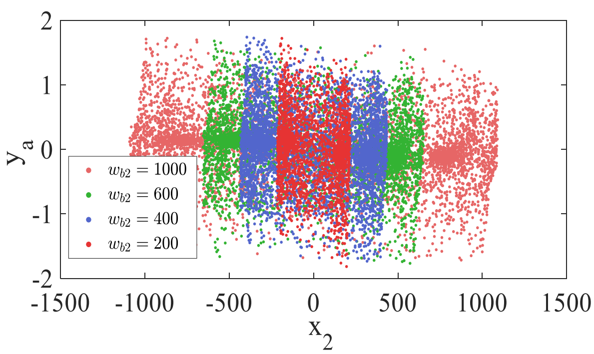

Fig. 18. Large-scale amplitude control of x_2 by a wide range of w_{b2}.

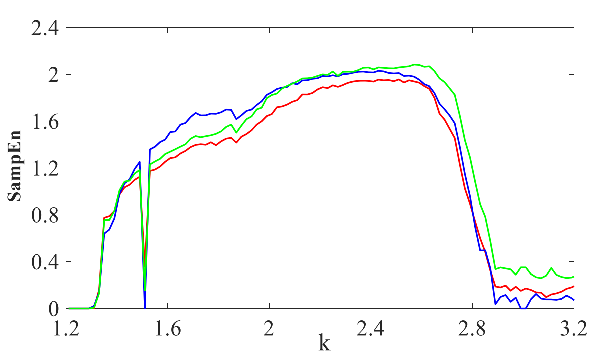

Fig. 19. Sample entropy of the DBMNN with respect to k: $x_2(n)$-red, $y_b(n)$-blue, $\varphi(n)$-green.

error between the two functions over the interval $[-4, 4]$ is 2.34%. Employing this approximation function $G(x)$ in FPGA-based chaotic circuit design improves hardware efficiency while preserving the original dynamical behavior.

The FPGA implementation of DBMNN is mainly composed of three parts: algorithm design, hardware planning, and experimental measurements. Fig. 21 shows the flow block diagram of the FPGA algorithm implementation for DBMNN, which comprises three modules: FPGA-based DBMNN, an intercept and make positive unit for processing data, and a DAC unit for converting digital signals into analog signals. The system

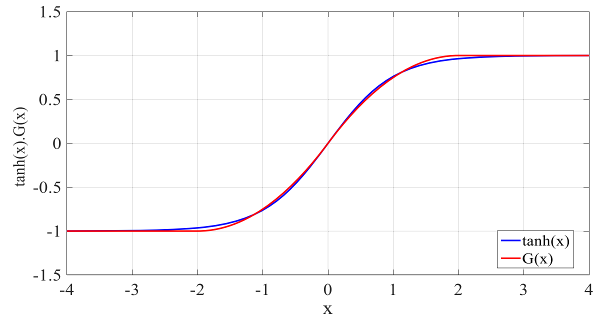

Fig. 20. Curves of the hyperbolic tangent function and the cecond-order piecewise function(L = 2).

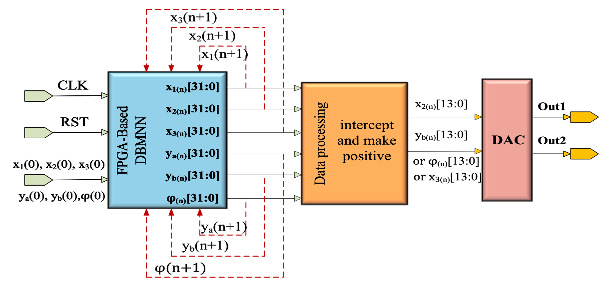

Fig. 21. Flow block diagram of FPGA-based DBMNN.

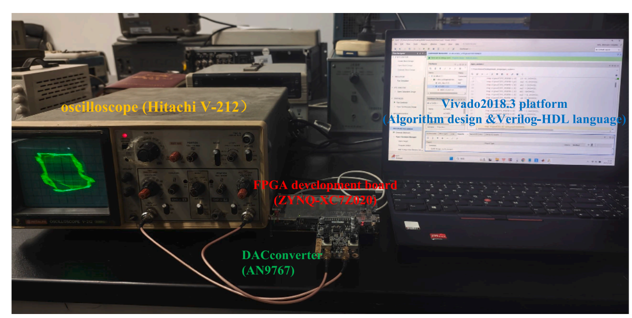

Fig. 22. Hardware devices composition for circuit implementation of DBMNN.

clock "CLK" and the reset signal "RST" are two 1-bit control input signals used for synchronizing each module unit. To ensure computational precision, a 32-bit fixed-point decimal format is employed, comprising 1 sign bit, 6 integer bits, and 25 fractional bits. The data bit-width processing in FPGA-based algorithm design mainly involves multiplication operations and truncation. Specifically, a 64-bit register is used to store the product of two 32-bit fixed-point numbers. Then, in the next state machine, the adapted 32-bit product is obtained by truncating the sign bit and the bits from positions 56 to 25 of the 64-bit temporary result.The input port receives initial values $(x_1(0), x_2(0), x_3(0), y_a(0), y_b(0), \varphi(0))$ as a 32-bit input signal. After computation by the first module, the 32-bit output signals $(x_1(n), x_2(n), x_3(n), y_a(n), y_b(n), \varphi(n))$ are fed back to the input port to for the next iteration and into the data processing block for DAC output signal preparation. The data processing module is responsible for adding appropriate offsets to the final 32-bit decimal data to make them positive and truncating bits [27 : 14]to obtain a 14-bit output consisting of 3 integer bits and 11 decimal bits. Finally, the DAC module transforms the digital signals into analog signals that can be presented on the oscilloscope.

The composition of the hardware equipment for the circuit experiment is shown in Fig. 22. Connecting the development board to both the analog oscilloscope and the PC data interface, we program the bitstream file generated by the Vivado platform into the development board. The

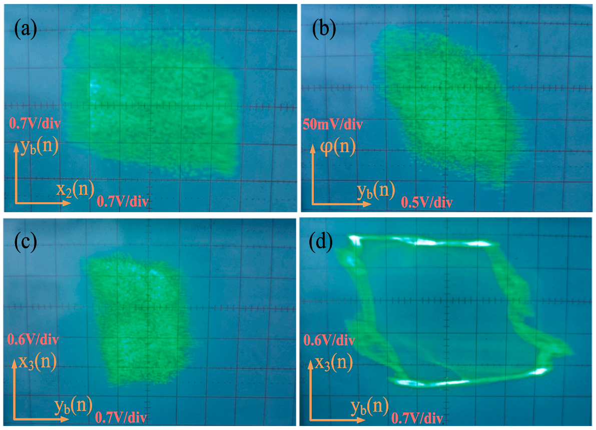

Fig. 23. Phase diagrams of the oscilloscope corresponding to DBMNN's chaotic attractors with different parameters.(a)$x_2 - y_b$ phase with $k = 2.5, w_{a1} = 2.5, w_{1b} = -3.2$. (b)$y_b - \varphi$ phase with $k = 2.5, w_{a1} = 2.45, w_{1b} = -3.2$. (c)$y_b - x_3$ phase with $k = 2.5, w_{a1} = 2.5, w_{1b} = -3.25$. (d)$y_b - x_3$ phase with $k = 2.5, w_{a1} = 2.5, w_{1b} = -2.04$.

Table 2
FPGA resource utilization of the DBMNN system.

Resource	Utilization	Available	Utilization %
LUT	2664	53,200	5.01
FF	1954	106,400	1.84
DSP	34	220	15.45
IO	34	125	27.20

phase diagrams of the DBMNN are then successfully displayed on the oscilloscope. The FPGA experimental results shown in Fig. 23(a), (b), (c), and (d) are consistent with the MATLAB simulation results in Figs. 8(c), 10(c), 12(b), and (d), respectively, thereby confirming the circuit feasibility of the proposed DBMNN. Table 2 presents the utilization rates of various FPGA resources. The results indicate that the FPGA resource utilization is acceptable, with the LUT, FF, DSP, and IO utilization rates at 5.01%, 1.84%, 15.45%, and 27.20%, respectively. This demonstrates that the FPGA-based implementation of the DBMNN model is efficient and suitable for practical deployment. Additionally, the simulated maximum power consumption is 67.18 W, and the calculated bit throughput is 0.3 Gbps.

5. Conclusion

In this paper, we conduct an innovative exploration of discrete neural network architectures capable of exhibiting chaotic dynamics, and propose a novel DBNN model with a bidirectional feedback structure inspired by the BAM neural network. To further enhance the nonlinear characteristics of DBNN, we embed a bistable locally active discrete memristor as a synaptic connection into its feedback path, constructing a six-dimensional DBMNN model. Compared with existing chaotic systems based on discrete memristive neural networks, the proposed DBMNN is the first to link a double-layer bidirectional network with a memristor and demonstrates complex dynamical behavior with a simple mathematical structure.

The influence of different control parameters on the dynamics of DBMNN is comprehensively analyzed using phase diagrams, Lyapunov exponent spectrum, bifurcation diagrams, and bi-parameter dynamic maps. Numerical simulations reveal that the system can exhibit extensive hyperchaos as well as various unique bifurcation phenomena under parameter variations, notably, its number of positive Lyapunov exponents characterizing hyperchaos can reach four. Under the effect of the bistable locally active memristor, the system can exhibit coexisting attractors of different dynamic properties for two initial conditions. Moreover, it is also found that DBMNN can exhibit interesting transient behaviors under specific parameters and initial conditions, including transient chaos and chaotic attractor jumping. Finally, the circuit design of the proposed DBMNN is successfully implemented on an FPGA hardware platform, and the experimental results are consistent with software simulations, verifying the physical feasibility of the system. This work offers a promising framework for brain-inspired modeling and neurocomputing by demonstrating hyperchaotic dynamics in the DBMNN system. Future works will further explore applications of DBMNN in fields such as image encryption and secure communication, as well as investigate novel discrete memristive neural network models with richer dynamical behavior.

CRediT authorship contribution statement

Chunhua Wang: Supervision, Funding acquisition, Conceptualization; **Zhibo Gong:** Writing – original draft, Methodology, Investigation; **Quanli Deng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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